

ST PETERSBURG PULKOVO, Russian Federation

WMO: 260630

Lat: 59.800N

Lon: 30.263E

Elev: 24

StdP: 101.04

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-22.9	-19.8	-25.9	0.4	-22.6	-22.5	0.5	-19.4	10.4	-0.8	9.4	-1.2	2.1	100	0.594

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	28.4	19.6	26.3	18.7	24.7	17.8	20.9	26.3	19.7	24.7	18.7	23.3	3.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.8	23.7	17.9	12.9	22.3	16.9	12.1	21.2	60.3	26.5	56.3	24.5	52.9	23.3	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.0	7.8	6.9	-25.9	31.2	4.0	2.3	-28.8	32.8	-31.1	34.2	-33.4	35.4	-36.3	37.1
			-26.1	22.9	4.0	1.3	-29.0	23.9	-31.3	24.7	-33.5	25.4	-36.4	26.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.7	-5.8	-6.1	-1.8	5.1	10.9	15.3	18.4	16.8	11.8	5.7	0.4	-3.3	(6)
(7)		DBStd	9.75	6.55	6.80	4.72	4.26	4.39	3.50	3.21	3.03	3.45	3.85	4.70	5.97	(7)
(8)		HDD10.0	2372	491	451	366	157	42	2	0	0	21	141	289	412	(8)
(9)		HDD18.3	4697	749	684	624	398	234	104	38	67	196	393	539	671	(9)
(10)		CDD10.0	793	0	0	0	9	69	161	261	210	76	7	0	0	(10)
(11)		CDD18.3	75	0	0	0	0	3	13	40	19	1	0	0	0	(11)
(12)		CDH23.3	681	0	0	0	2	39	131	346	158	5	0	0	0	(12)
(13)		CDH26.7	156	0	0	0	0	7	29	89	32	0	0	0	0	(13)
(14)	Wind	WSAvg	3.1	3.4	3.4	3.4	3.1	3.0	2.9	2.6	2.7	2.8	3.2	3.4	3.6	(14)
(15)	Precipitation	PrecAvg	650	45	35	34	34	50	70	80	82	57	60	58	47	(15)
(16)		PrecMax	846	78	88	61	74	122	141	161	184	156	107	113	92	(16)
(17)		PrecMin	470	17	3	8	6	11	26	12	24	15	21	0	14	(17)
(18)		PrecStd	100	15	18	14	18	25	27	36	41	37	23	26	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	4.9	11.9	22.1	27.2	29.8	31.8	29.8	23.9	16.2	10.1	8.2	(19)
(20)			MCWB	3.9	3.3	6.2	13.3	17.6	20.1	21.4	19.8	17.4	12.7	8.8	7.1	(20)
(21)		2%	DB	3.2	3.8	7.7	18.1	23.9	27.0	29.1	27.1	21.0	14.0	8.7	6.1	(21)
(22)			MCWB	2.5	2.5	4.0	10.7	15.7	18.9	20.3	18.8	16.1	11.8	7.7	5.2	(22)
(23)		5%	DB	2.2	2.8	5.2	15.0	21.2	24.2	27.0	24.9	18.9	12.3	7.7	4.8	(23)
(24)			MCWB	1.6	1.8	2.6	9.0	13.7	17.2	19.6	18.1	14.6	10.6	6.8	4.1	(24)
(25)		10%	DB	1.6	1.9	3.9	12.2	19.0	22.1	25.0	23.0	17.1	11.2	6.2	3.3	(25)
(26)			MCWB	1.1	1.1	2.0	7.5	12.5	15.8	18.5	17.2	14.0	9.6	5.5	2.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	3.8	6.5	13.8	19.4	21.2	22.7	21.8	18.3	13.3	9.0	7.1	(27)
(28)			MCD B	4.3	4.4	10.5	21.1	25.6	27.4	29.0	27.6	22.3	15.4	9.7	8.0	(28)
(29)		2%	WB	2.6	2.6	4.3	11.5	16.7	19.7	21.4	19.7	16.6	12.1	7.6	5.1	(29)
(30)			MCD B	3.0	3.2	6.6	16.7	21.5	25.3	27.1	24.4	19.8	13.7	8.3	5.8	(30)
(31)		5%	WB	1.7	1.7	3.1	9.7	14.8	18.2	20.4	18.7	15.4	11.1	6.6	3.9	(31)
(32)			MCD B	2.1	2.4	4.7	14.2	19.7	23.1	25.4	23.5	18.1	12.5	7.3	4.4	(32)
(33)		10%	WB	0.9	1.0	2.2	7.8	13.1	16.7	19.3	17.8	14.4	10.0	5.5	2.6	(33)
(34)			MCD B	1.3	1.6	3.4	11.3	18.1	20.8	24.0	22.1	16.7	11.1	6.2	3.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.4	6.3	7.6	9.3	10.5	9.8	9.7	9.4	8.1	5.7	4.5	4.9	(35)
(36)			MCD BR	3.5	4.4	9.0	13.9	14.3	13.4	12.9	12.7	10.7	7.2	4.7	4.0	(36)
(37)			MCWBR	3.4	4.0	6.2	8.1	7.7	7.0	6.1	6.1	6.4	5.3	4.4	3.7	(37)
(38)		5% WB	MCD BR	3.5	4.4	7.3	12.7	12.6	11.8	11.6	11.0	9.1	6.6	4.4	4.0	(38)
(39)			MCWBR	3.5	4.2	5.4	7.8	7.4	6.7	6.1	5.9	6.0	5.2	4.3	3.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.277	0.315	0.340	0.380	0.374	0.370	0.396	0.390	0.358	0.336	0.304	0.262	(40)	
(41)		taud	2.116	2.130	2.192	2.277	2.379	2.420	2.367	2.376	2.450	2.384	2.209	2.082	(41)	
(42)		Ebn at Noon	484	664	779	817	854	863	826	801	768	656	453	354	(42)	
(43)		Edh at Noon	45	78	100	110	107	105	108	100	79	62	42	31	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.27	0.88	2.34	3.75	5.09	5.33	5.15	4.04	2.45	0.99	0.28	0.15	(44)	
(45)		RadStd	0.05	0.20	0.31	0.45	0.51	0.39	0.58	0.43	0.23	0.15	0.04	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.68	N/A	N/A	N/A	N/A	N/A	-242	-283	N/A	N/A
(47) Regional (15 neighbors)		+0.52	N/A	N/A	N/A	N/A	N/A	-182	-180	N/A	N/A

Nomenclature: See separate page